

MODULE 4

Design Engineering Concepts

SYLLABUS

- Design Engineering Concepts:-Project-based Learning and Problem-based Learning in Design.
- Modular Design and Life Cycle Design Approaches.
- Application of Biomimicry, Aesthetics and Ergonomics in Design.
- Value Engineering, Concurrent Engineering, and Reverse Engineering in Design.

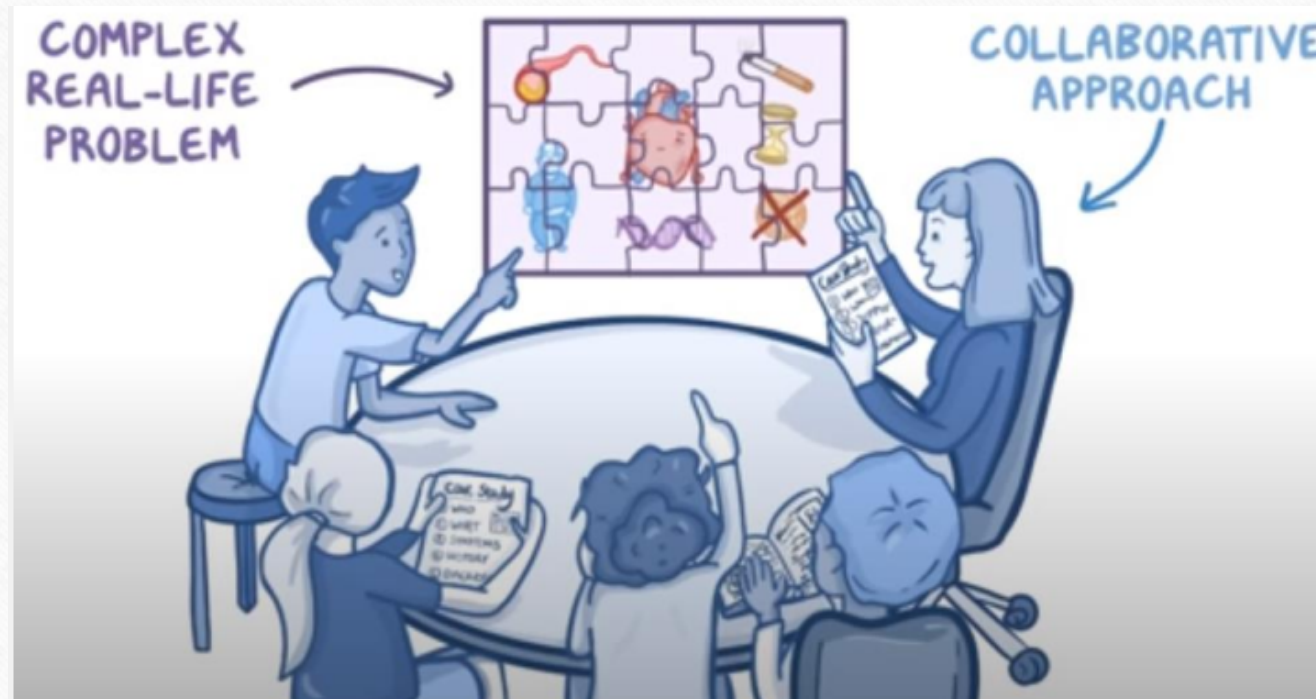
PROBLEM-BASED LEARNING

- It empowers learners to conduct research, integrate theory and practice, and apply knowledge and skills to develop a viable solution to a defined problem.
- It is a teaching pedagogy that is student- centered
- Students learn about a topic through the solving of problems and generally work in groups to solve the problem where, often, there is no one correct answer.

PROBLEM-BASED LEARNING

- Problem-based learning typically follow prescribed steps:
 1. Presentation of an "ill-structured" (open-ended, "messy") problem
 2. Problem definition or formulation (the problem statement)
 3. Generation of a "knowledge inventory" (a list of "what we know about the problem" and "what we need to know")
 4. Generation of possible solutions
 5. Formulation of learning issues for self-directed and coached learning
 6. Sharing of findings and solutions

PROBLEM-BASED LEARNING (PBL)



PBL - EXAMPLE

WHERE is the PAIN ?

WHEN did it START ?

OTHER SYMPTOMS ?

MEDICAL/ FAMILY HISTORY ?

OTHER PROBLEMS ?

72 yrs old MAN

CHEST

ANATOMY OF THE HEART

ECG

Cardiology

ESSENTIALS

Myocardial infarction

HEART ATTACKS AND PREVENTION

34 19

The collage consists of several elements: a vertical list of five questions in purple speech bubbles on the left; a central cartoon of a 72-year-old man in a blue shirt clutching his chest, with the word 'CHEST' and an arrow pointing to his hand; and a collection of medical illustrations on the right, including an ECG graph, a test tube, a heart anatomy diagram, a chest X-ray, a 'Cardiology' label, a 'Myocardial infarction' diagram, and a 'HEART ATTACKS AND PREVENTION' graphic with a red heart and a person running.

PBL - EXAMPLE



PBL - EXAMPLE

→ AT the END :

Learning issues:

* How does the heart pump blood ?

○

* Why do heart attacks cause pain ?

○



DIVIDE up &
RESEARCH on
their OWN

PBL - EXAMPLE

BEFORE the NEXT CLASS :

STUDENT 1

STUDENT 2

STUDENT 3

Learning issues:

* How does the heart pump blood ?

- OSMOSIS video: CARDIOVASCULAR ANATOMY & PHYSIOLOGY
Refer to lecture 18 slides
Textbook

- * Why do heart attacks cause pain ?

www.OSMOSIS.ORG/MYOCARDIAL INFARCTION
Textbook

STUDENT 4

STUDENT 5

STUDENT 6

PBL - EXAMPLE



ACTIVE ENGAGE IN LEARNING

- ✓ MEMORIZE CONTENT
- ✓ WORK TOGETHER
- ✓ BUILD A STRONG WEB OF INTER CONNECTED INFORMATION
- ✓ SOLVE REAL WORLD PROBLEMS

PROJECT-BASED LEARNING

- Project-based learning is an instructional approach where we learn by investigating a complex question, problem or challenge.
- It promotes active learning, engages students, and allows for higher order thinking
- Students explore real-world problems and find answers through the completion of a project.
- Students also have some control over the project they will be working on, how the project will finish, as well as the end product.

PROJECT-BASED LEARNING

- Involves
 - Knowledge
 - Critical thinking
 - Collaboration
 - Communication

DIFFERENCE BETWEEN PROBLEM BASED AND PROJECT BASED LEARNING

- Students who complete problem-based learning often share the outcomes and jointly set the learning goals and outcomes with the teacher.
- On the other hand, project-based learning is an approach where the goals are set. It is also quite structured in the way that the teaching occurs.

DIFFERENCE

PROJECT BASED LEARNING

- goals are set and quite structured
- often multidisciplinary and longer
- follows general steps
- involves authentic tasks that solve real world problems

PROBLEM BASED LEARNING

- often share the outcomes and jointly set the learning goals and outcomes
- more likely to be a single subject and shorter

PROJECT BASED LEARNING VS PROBLEM BASED LEARNING

Similarities

Both PBLs:

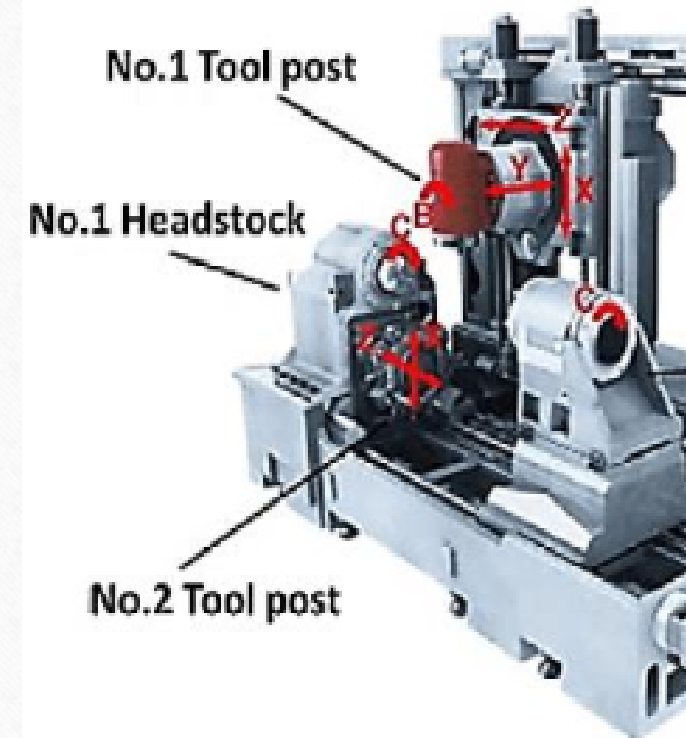
- Focus on an open-ended question or task
- Provide authentic applications of content and skills
- Build 21st century success skills
- Emphasize student independence and inquiry
- Are longer and more multifaceted than traditional lessons or assignments

Differences

Project Based Learning	Problem Based Learning
Often multi-subject	More often single-subject, but can be multi-subject
May be lengthy (weeks or months)	Tend to be shorter, but can be lengthy
Follows general, variously-named steps	Classically follows specific, traditionally prescribed steps
Includes the creation of a product or performance	The "product" may be tangible OR a proposed solution, expressed in writing or in a presentation
May use scenarios but often involves real-world, fully authentic tasks and settings	Often uses case studies or fictitious scenarios as "ill-structured problems"

MODULAR DESIGN

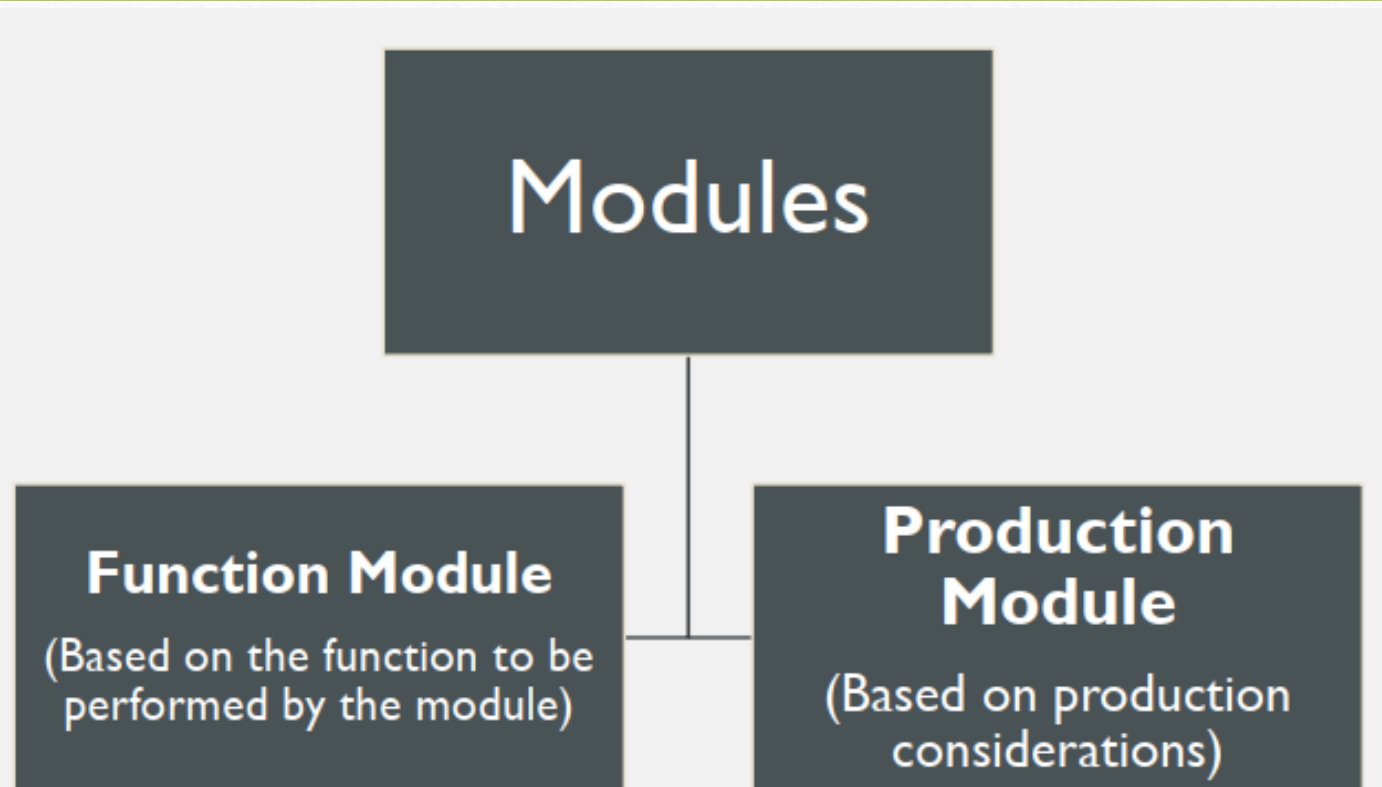
- Module' means separate elements
- Modular design is an approach in which a product is designed for assembling in module-wise fashion.
- Modular products are the artifacts that are composed of many modules
- These modules function together to get the overall function of the product.
- Modular products can be machines, assemblies and components that fulfill various overall functions through the combination of distinct building blocks or modules.
- In a modular product (or modular system), the overall function performed by the product is the results achieved through a combination of discrete units (modules).



MODULARIZATION

- Dividing a product into discrete units based on some criteria is called as modularization of a product.
- As we have seen, modular products or modular Systems are built up on separable or inseparable units called as modules.
- The basic idea behind modular design is to organize a complex system as a set of distinct component that can be developed independently and then assembled together to perform a function.

MODULARIZATION



MODULAR DESIGN PROCESS

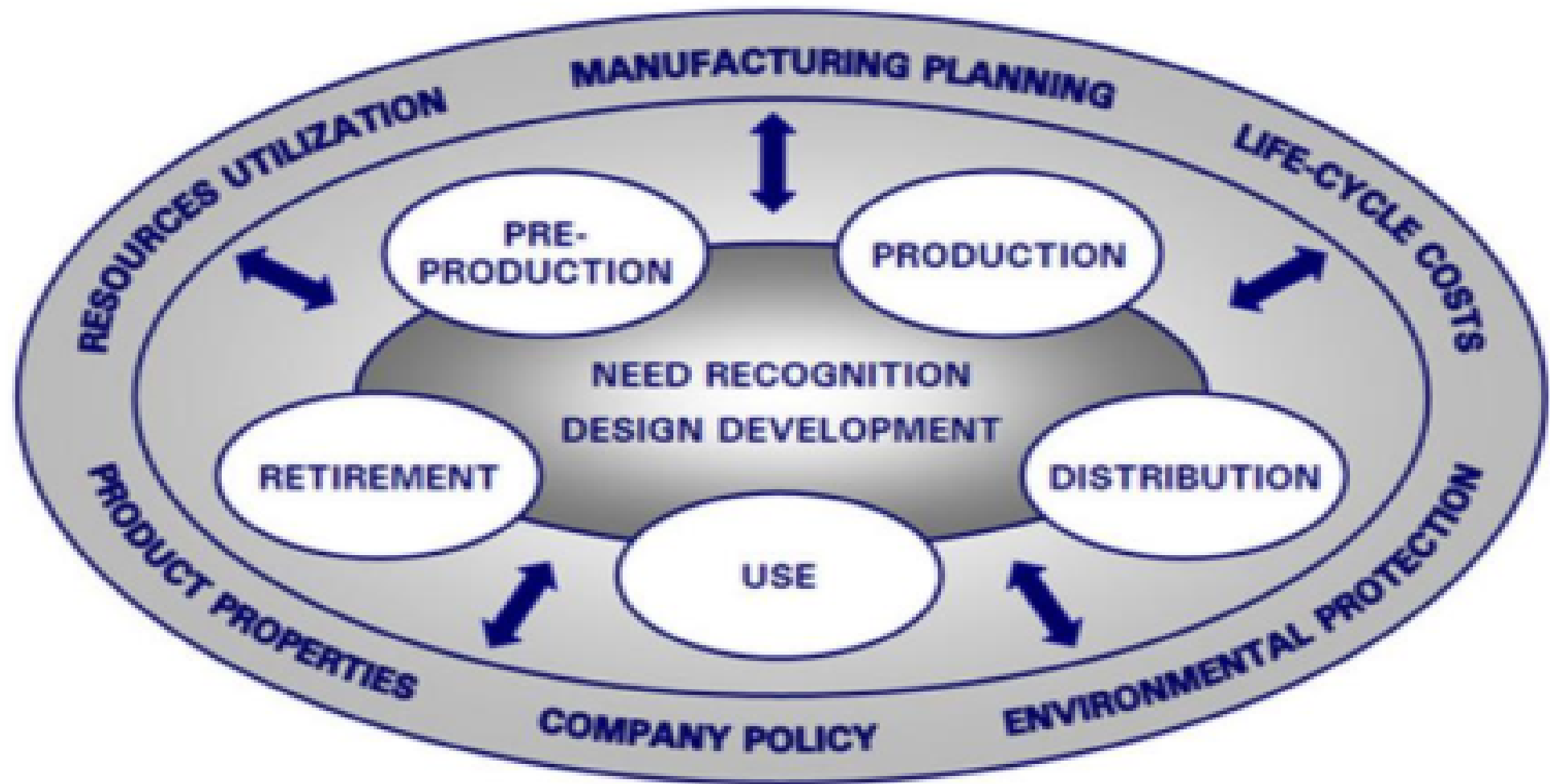
- We give importance to designing of module rather than the product as a whole
- Stage 1 : clarify the task
- Stage 2 : establish function structure
- Stage 3 : searching for solution principles and concept variants
- Stage 4 : selecting and evaluating
- Stage 5 : Preparing design and dimensioned layouts
- Stage 6 : Preparing production document

ADVANTAGES OF MODULAR DESIGN

- Minimizing cost
- Design of a single part is easier as designer can concentrate only in one section
- Module can be separately improved without affecting the entire product
- A part of module can be updated
- Replacement of a part becomes cheaper
- Shorten the design cycle
- Improves reliability and quality

LIFE CYCLE DESIGN

- The application of the life cycle concept to the design phase of the product development process is known as Life Cycle Design (LCD)
- a design intervention which takes into consideration all the phases of a product's life cycle
 - Development
 - Production
 - Distribution
 - Use
 - Maintenance
 - Disposal
 - Recovery



LIFE CYCLE DESIGN

- As a design approach, Life Cycle Design is characterized by three main aspects:
 - the perspective broadened to include the entire life cycle;
 - the assumption that the most effective interventions are those made in the first phases of design;
 - the simultaneousness of the operations of analysis and synthesis on the various aspects of the design problem.
- Main phases of a product's life cycle
 - Recognition and design development
 - Pre-production
 - Production
 - Distribution
 - Use
 - Retirement,

LIFE CYCLE DESIGN

- The selection of design alternatives must be guided by considering the main factors of product success (design targets), in relation to all the phases of the life cycle:
 - Resources utilization
 - Manufacturing planning
 - Life cycle cost
 - Product properties (ease of production, functionality, safety, quality, reliability, aesthetics)
 - Company policies
 - Environmental protection.

ERGONOMICS IN DESIGN



- The word derived from Greek, 'ergon' means work and 'nomos' means laws
- Ergonomics is basically the science of analyzing work and then designing items (tools, equipment, products) and methods to most appropriately fit the capabilities of the user.
- Ergonomics design approach focuses on human comfort and decreased fatigue through product design.
- Means, during the design phase of a product, all the aspects of the product that can cause discomfort while using that product are identified. Then, analyzes the causes of the discomfort and appropriate solutions will be incorporated in the product design



ERGONOMICS IN DESIGN

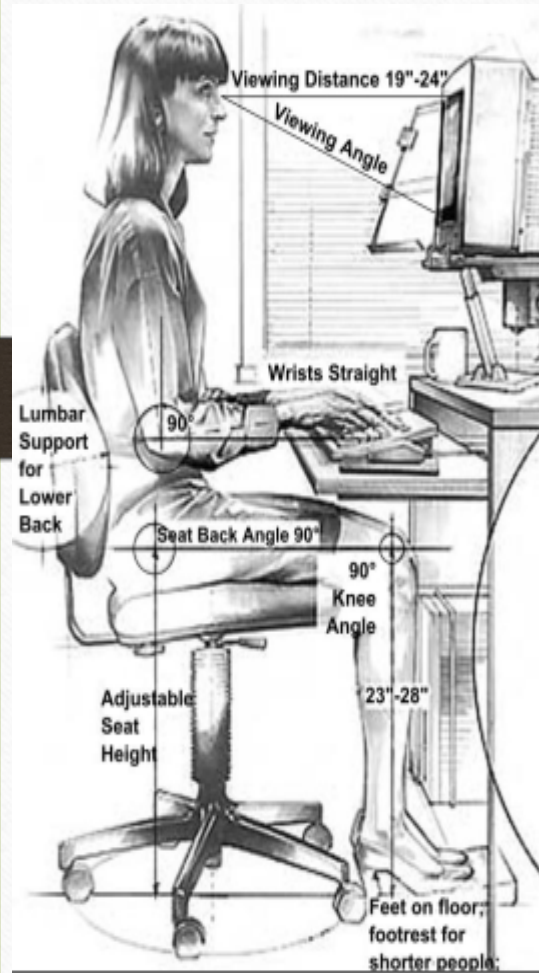
- Ergonomic design applied to an office chair will focus on how much it is comfortable for a Person who sits on it during office work.
- A chair ergonomically designed for dining purpose and a chair meant for relaxed sitting at beach will be different.
- It is, because, the kind of comfort and function to be provided by the chairs in these situations are different

FACTORS CONSIDERED IN ERGONOMICS DESIGN

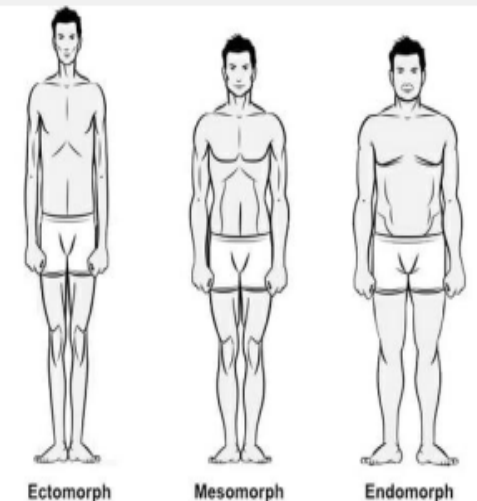
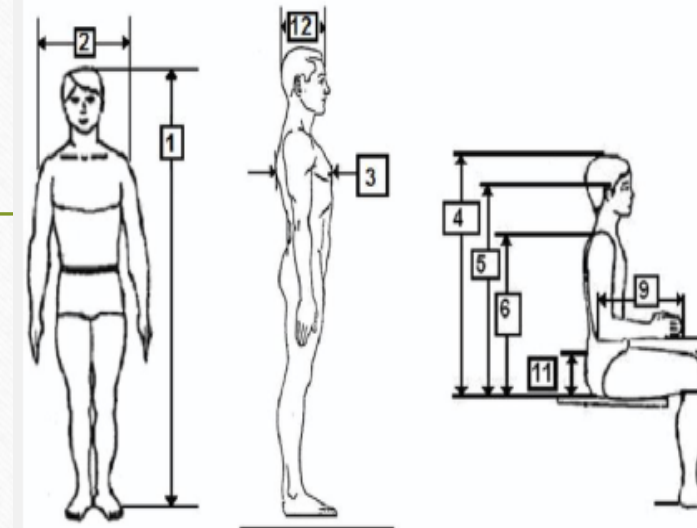
- To develop an ergonomic design (for a product or system), the designer will have to consider and analyze anthropometric data (dimensions of human body), posture of working while using the product, kind of movements and kind of workspace.
- ultimately, ergonomic design involves every aspect of user-product interaction, for the comfortable utilization of a product.



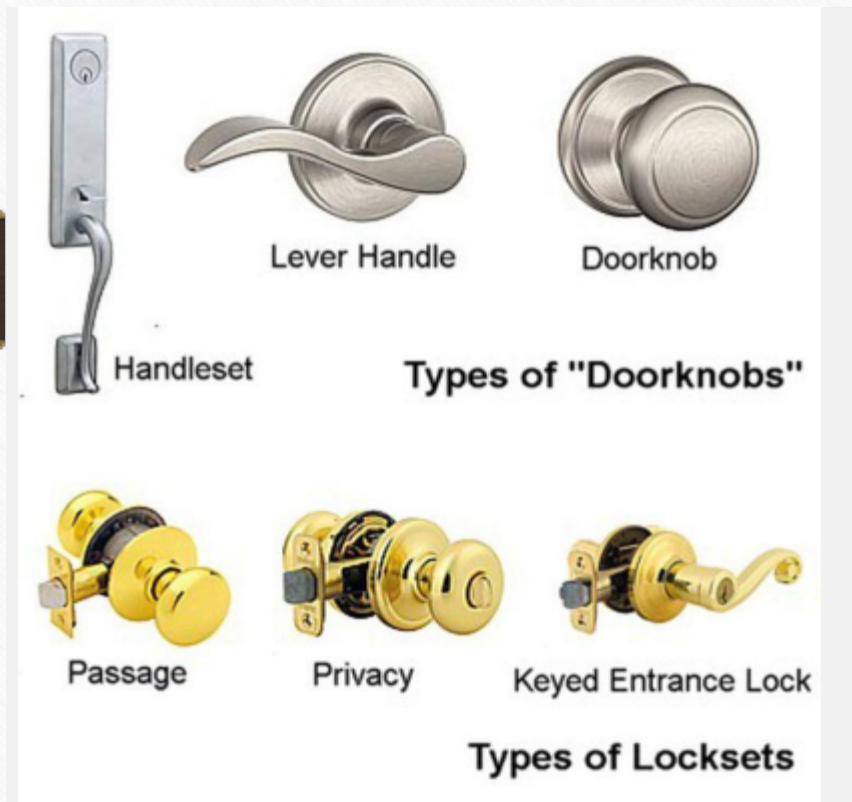
ANTHROPOMETRY



- Anthropometry is the science that measures the range of body sizes in a population.
- While designing products, it is very important for a designer to remember that people come in many sizes and shape.
- The anthropometric data vary considerably between human races.
- Age and occupation of the user is also relevant in anthropometric study.
- Working posture is another important factor to be considered in ergonomic design. Posture can be standing, sitting, reaching, moving and combinations of any of these (such as sitting and moving, standing and reaching, etc.).



UNIVERSAL DESIGN CONCEPT



- The designer must recognize the special needs of different users, including the individuals with disabilities.
- The disability can be temporary or permanent.
- Few examples for disabilities include broken bones, sprained joints, pregnancy, handicap, differently able, aging, etc.
- Universal design is an approach in ergonomics design, which considers all people; common and special people; who are potential users of a product.

ADVANTAGES OF ERGONOMIC DESIGN

- Proper consideration of ergonomic design can bring lots of advantages in working environment
- health issues can be solved
- Increase savings because of productive, sustainable and effective work environment
- Reduce medical expenditure

AESTHETICS IN DESIGN

- The word 'aesthetics' is derived from the Greek word 'aesthetikos' meaning sensory perception.
- Aesthetics is the feel that a human being perceives.
- When a person perceives a sense of pleasure through any of the senses while using a product, then we can say that the product is aesthetically appealing.
- Example: a beautiful person, a good food, nice perfume

AESTHETICS IN ENGINEERING

- Products are intentionally designed to generate a defined perception in potential customers
-
- Aesthetics of a product (that is how a customer feels about a product) is a very important aspect for its business merit and acceptability.
 - This feel (or perception) enables the customer to distinguish and choose a product from similar products.
 - Few examples for demarcation of perceptions are; hot and cold, smooth and rough, soft and hard, heavy and light, dark and bright, sweet and sour, loud and quiet, sharp and dull, spacious and congested, etc.. customers generally combine few of these feels (or attributes) and arrive at conclusion of a product as reliable, enjoyable and precise.

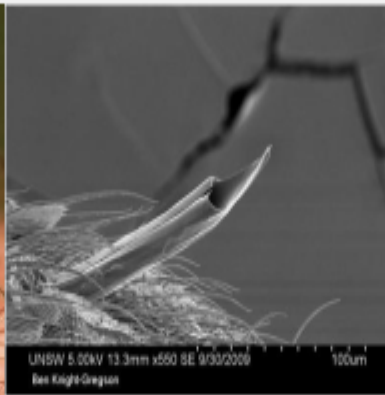
AESTHETICS IN DESIGN

- The customer may not know why a product is aesthetically appealing for them. it can be appealing because the designer has incorporated some specific attributes the design in order to create such a feel in the customer.



BIO-MIMICRY IN DESIGN

- “Biomimicry borrows nature's blueprints, recipes, processes, and ecosystem strategies and then comes up with design principles to solve our own problems



Termite-Inspired Air Conditioning



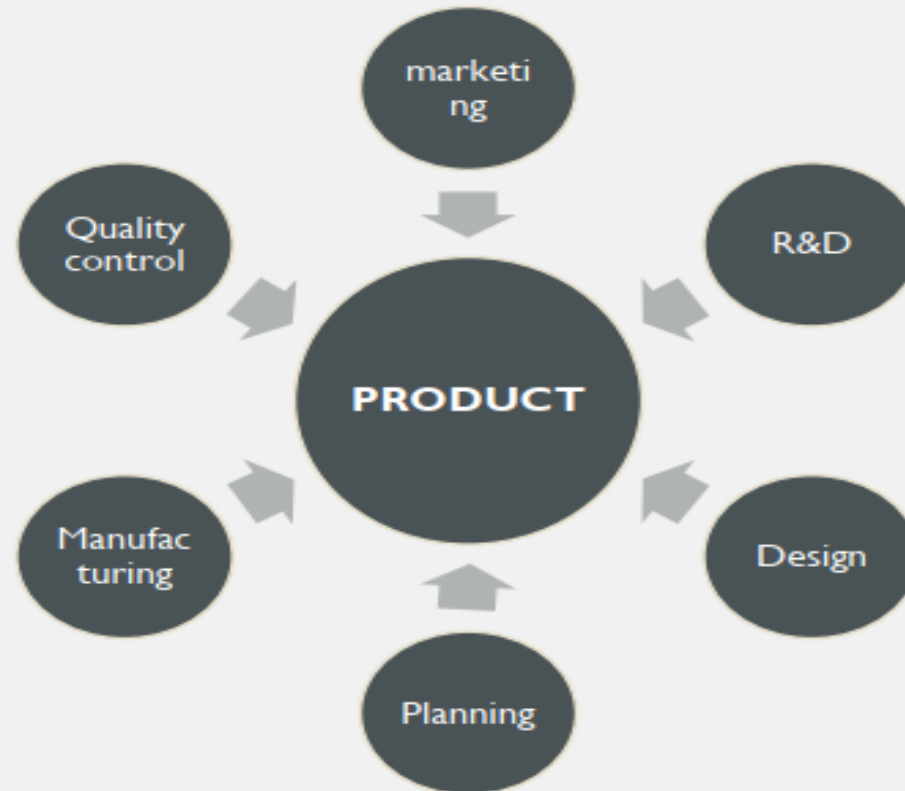
Eastgate (Harare, Zimbabwe) uses only 10 percent of the energy of a conventional building its size, **saved 3.5 million in air conditioning costs** in five years, and has rents that are 20% lower than a newer building next door.



CONCURRENT ENGINEERING

- It is an approach in product design process in which people from various functional areas works together simultaneously to develop a product.
- Since people from various fields are working simultaneously for the development, this kind of engineering is also known as Simultaneous Engineering or Parallel Engineering.
- This approach is adopted to improve the efficiency of product design and reduce the product development cycle time.

CONCURRENT ENGINEERING



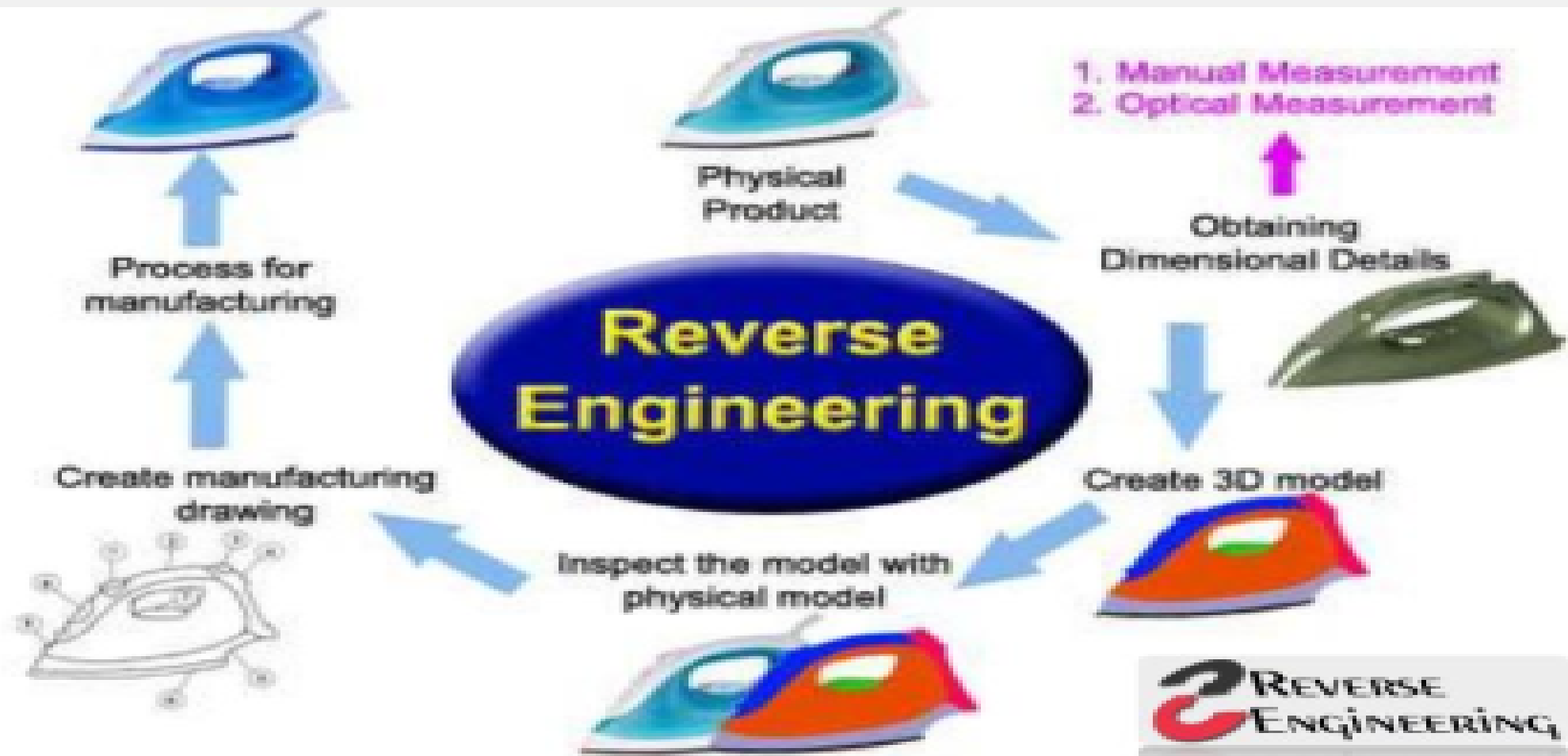
ADVANTAGES OF CONCURRENT ENGINEERING

- Reduce design time
- Reduce manufacturing
- Reduce cost for design changes
- Ensure correct data and information transfer between various sections
- Simultaneous thinking leads to amazing innovations
- Every person has feel of belongingness to the product

REVERSE ENGINEERING

- Reverse Engineering is an approach in which an existing product is analyzed and another product is developed in light of the analysis.
- The product that is analyzed can be own product of the producer or a product from a competitor.
- In reverse engineering, a product is dissected or dis-assembled to find out in detail how a part works and why it is used. This information obtained by this process can then be applied to solve own design problem or develop a new product.
- Reverse Engineering is essentially a functional decomposition process in the reverse direction.
- an existing product is analyzed into subsystems, which are further analyzed in depth to ultimately establish the product concept
- This analysis will help the designer to identify weak side of the design

REVERSE ENGINEERING



Value Engineering

What is Value Engineering?

- Value Engineering refers to the systematic method of improving the value of a product that a project produces.
- It is used to analyze a service, system, or product to determine the best way to manage the important functions while reducing the cost.

Value Engineering

- Value Engineering encourages using alternative methods and materials that are less expensive and do not lower the functionality of the system, service, or product.
- Value engineering refers to a systematic approach to improve the value of a project at the lowest cost.
- The concept of Value Engineering began in the 1940s at General Electric.

Value Engineering

- **Understanding value engineering**

- Value is calculated as the ratio of function to the cost
- $\text{Value} = \text{function} / \text{cost}$
- A business can add value to a product by either cutting down on cost or improving the function.
- Most companies use value engineering as the cost – cutting strategy, where the basic function of a product is preserved – not sacrificed – in the process of pursuing value improvement.

Value Engineering

- The manufacturer uses the cheaper components that meet the product lifetime duration while preserving the basic purpose of the product.

History of Value Engineering

- The concept of value engineering started in the 1940s at General Electric Co.
- This was during the Second World War when there was a shortage of raw materials, component parts, and skilled labor.
- The engineers at General Electric had to find alternative components and raw materials to ensure the continuity of the production process.
- Therefore, Lawrence Miles, Harry Erlicher, Jerry Leftow, and other engineers sourced for acceptable substitutes that would reduce the production costs without compromising the functionality of the products.
- This concept turned into a systematic process that not only reduced the cost of production but also provided better final products or better performance.
- The engineers named this technique “value analysis.”

Steps in Value Engineering

- Value engineering can be broken down into the following phases:
 1. Information
 2. Function Analysis
 3. Creative
 4. Evaluation
 5. Development
 6. Presentation
 7. Implementation

Steps in Value Engineering

1. Information

- The information phase involves gathering project information and refining the goals of the project.
- Data is collected and analyzed, and the information obtained is used to finalize the priorities of the project and areas of improvement.
- The potential issues are broken down into constituent components, which are elements to be addressed.
- This phase also involves identifying the methods that the team will use to evaluate the progress of the project.

Steps in Value Engineering

2. Function Analysis

- The function analysis phase involves determining the functions of the project and identifying them with a verb/noun combination for every element under evaluation.
- The function is defined as the set targets to be attained through the execution of an element or a set of elements.
- Each of the identified functions is analyzed to determine if there are improvements to be made and if a new function is required.
- A cost is assigned to each identified function.

Steps in Value Engineering

3. Creative

➤ The creative phase follows the function analysis phase, and it involves exploring the various ways to perform the functions identified in the function analysis phase.

➤ This allows team members to brainstorm alternatives to existing systems or methods that are in use.

➤ Brainstorming forces people to be creative and allows team members to speculate on all possible solutions to the problems presented, or alternatives to the function.

➤ The team is required to develop a list of potential solutions to the function formulated by the verb/noun combination.

Steps in Value Engineering

4. Evaluation

- In the evaluation phase, the merits and demerits of each of the suggested solutions and alternatives from the creative phase are listed.
- The team should describe each advantage and disadvantage in general terms.
- When the disadvantages exceed the advantages, the alternative is dropped in favor of other solid alternatives.
- The team performs a weighted matrix analysis to group and rank the alternatives, and the best alternatives are selected for consideration in the next phase

Steps in Value Engineering

5. Development

- The development phase involves conducting an in-depth analysis of each best alternative to determine how it can be implemented and the cost involved.
- The examination of each alternative may involve creating sketches, cost estimates, and other technical analysis.
- Team members formulate an implementation plan for the project, which describes the process to be followed in implementing the final recommendations.

Steps in Value Engineering

6. Presentation

- The presentation phase is where the team meets with the management and other stakeholders to present their final report.
- The team is required to present their findings to the decision-makers using reports, flow charts, and other presentation materials to convince them that the final ideas from the development phase should be implemented.
- The ideas should be described in detail, including associated costs, benefits, and potential challenges.
- The final report acts as a record of the team's accomplishments during the study and a summary of the team's findings.
- It can also act as a reference tool for the company in future projects.

Steps in Value Engineering

7. Implementation

- Implementation of the project begins after the management's approval of the team recommendations.
- If there are changes requested by the management or other decision-makers, these changes should be incorporated into the implementation plan before the implementation begins.
- When implementing the project, the team should ensure that the primary goal of increasing value is achieved.
- The actual cost savings of the project should be determined based on the implementation of the recommendations.

ASSIGNMENTS – PROBLEM BASED LEARNING

- Design a new waste management system in your residence area, especially for plastic waste
- How can you market your own product within limited budget
- Plan your relative's wedding by following covid protocol

ASSIGNMENT – PROJECT BASED LEARNING

- Analyzing the five most popular social media platforms for teens, then predict and design a new platform based on existing trends and past trajectory of change.
- Solving the problem of negative and/or ‘fake news.’
- Imagine and discuss college education system in 2050

EXERCISE

- Ergonomically design a vegetable knife for your kitchen, consider gripping material, shape, safety and placement of knife

ACTIVITY

- Objective : To propose a new design for screw driver based on reverse engineering method.
- Design requirement : conventionally for different screws different heads are available. Present requirement is to develop a screw driver that can handle any screws without changing the heads. You can change the designs as per your wish.

